

Another major issue with such simple keyword based approaches is their inability to comprehend context. This can lead to a large number of false positives, wherein posts which are objecting to hate speech are marked spuriously as 'hate speech'. For instance, consider this sentence: "I wish people wouldn't call me nigger." This gets marked as objectionable content by a simple lexicon based approach, whereas the actual post is objecting to racist content. Similarly, simple keyword based approaches cannot understand literary quotes which contain such words. For instance, if someone is using a quote from the book *To Kill a Mockingbird* which uses the word 'nigger', that post should not be flagged as racist. Given the simplicity of lexicon based approaches, they cannot handle nuanced meanings. And similar problems occur in video and speech content moderation as well; so content moderation requires more sophisticated approaches.

An obvious extension of simple keyword based approaches is to consider longer phrases to identify objectionable content. But even with longer phrases, it is quite easy to end up with false positives and false negatives. We need approaches that can understand the context in which an utterance is being made, and to use the contextual information, to identify the following:

- Whether an objectionable word has actually been used with the intention to hurt or harm
- Whether sentences that do not contain any obvious objectionable words are still intentionally hurtful or meant to cause harm

An alternative to the keyword based approach is to use machine learning based techniques to detect objectionable content. Here, the idea is to train classifiers that can look at labelled objectionable content and learn features that can classify unlabelled content as objectionable or not. The features can be as simple as n-grams, or be more complicated, such as entity, sentiment, emotion, etc. However, this approach requires extensive 'hate speech' annotated data, which is hard to come by.

In order to overcome this issue, researchers have been looking for weakly supervised approaches for hate speech recognition. One such approach was used by researcher Haji Mohammad Saleem and his team in a paper titled, "A web of hate: Tackling hate speech in online social spaces," available at <https://arxiv.org/pdf/1709.10159.pdf>. The team identified online communities that are potential sources for hate speech and used content from those communities as labelled data for hate speech. Team members then trained a number of classifiers using this data and were able to classify text as hate speech or not.

By making the assumption that certain communities tend to indulge in hate speech, the team was able to overcome the requirement for large quantities of labelled data to train a supervised training model. However, the disadvantage of this approach is that not all speech from such a community is objectionable; hence this can lead to a number of false positives. Also, this system fails to catch certain simple cases of hate speech if the content is not similar to that earlier seen on these online forums. For instance, their system fails to classify the sentence "Black people are terrible" as hate speech, which would have been caught by an intelligent lexicon based approach.

When detecting objectionable content or hate speech, it is also important to identify the target of the hate speech. For instance, it is important to know whether the hate speech is directed against a specific individual, which could be the case with cyber bullying in community forums; or against a specific entity, like the US government or the LGBT community. This requires targeted entity identification for hate speech, which makes the task more challenging.

Another important variant is explicit hate speech versus implicit hate speech. Explicit hate speech is easy to detect using lexicon based approaches, whereas implicit hate speech is more amenable to machine learning based techniques. For instance, consider the following example: "Totally fed up with the way this country has turned into a haven for terrorists. Send them all back home." Note that this is an example of hate speech directed against a specific community, namely, immigrants. There are no explicit keywords for hate speech in this example; hence lexicon based approaches will not be able to detect this. Given the short passage that reveals limited context, this example becomes difficult even for context based approaches. To detect such implicit nuances of hate speech requires the involvement of humans. This is another case for which a fully automatic NLP/ML approach is not feasible, because of the inherent ambiguities in a language. Hence, we still need human moderators to classify the ambiguous cases. The main benefit of NLP/ML based approaches is to reduce the load on the human moderators, but not to do away with the latter entirely.

I also wanted to direct our readers to an excellent article on the role of artificial intelligence by Prof. Michael Jordan available at: <https://medium.com/@mijordan3/artificial-intelligence-the-revolution-hasnt-happened-yet-5e1d5812e1e7>. He talks about how, instead of AI, it is intelligence augmentation (machines augmenting human intelligence) that is the driving force behind much of the so called AI revolution. He clearly points out that human imitative AI or what is referred to as Artificial General Intelligence is still far from real and it is the pursuit of Intelligence Augmentation that is the immediate way forward. This is definitely an important article to read if you are interested in the future roadmap of AI.

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Wishing all our readers happy coding until next month! **END** 🐧

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